

Anamitra Majumder

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Objective

Driven and analytically grounded Data Science graduate student at IIIT Lucknow, with a solid foundation in mathematics and computing, and a specialization in Quantum AI and statistical modeling. Passionate about leveraging data to solve complex problems through the integration of machine learning, quantum computing, and big data systems.

Education

Indian Institute of Information Technology, Lucknow

M.Sc. in Data Science Ongoing
CGPA: 8.2/10

Key Modules: Machine Learning, Deep Learning, Data Structures and Algorithms, Advanced Statistics

Achievements: JAM 2024 (Statistics), AIR 173; TOEFL: 108/120 (June 2024)

Institute of Mathematics and Applications

B.Sc. in Mathematics and Computing 2023
CGPA: 7.10/10

Delhi Public School, Ruby Park

Class 12 CBSE	2020	Score: 92.6%
Class 10 CBSE	2018	Score: 92.6%

Technical Skills

- **Programming Languages:** Python, C, Binary
- **Machine Learning & Statistical Analysis:** Scikit-learn, Statsmodels, XGBoost, PyCaret, TensorFlow, PyTorch
- **Data Manipulation & Visualization:** NumPy, Pandas, Matplotlib, Seaborn, Plotly, Tableau
- **Big Data Tools:** Apache Spark (PySpark), Hadoop, Hive, Kafka, Dask
- **Databases:** SQL, PostgreSQL, MongoDB
- **Quantum Computing & AI:** Qiskit, PennyLane, TensorFlow Quantum, TorchQuantum, Quantum Neural Networks (QNNs), Quantum Monte Carlo
- **Web & Conversational AI:** Flask, Rasa, Streamlit
- **Finance & Quantitative Modeling:** Time-series modeling, Financial sentiment analysis, QGANs, Statistical forecasting, Monte Carlo simulation

Soft Skills

- Communication – clear articulation of complex technical concepts
- Teamwork – successful collaboration with interdisciplinary teams
- Problem-solving – strong analytical thinking and structured reasoning

- Adaptability – quick learner in dynamic, high-pressure environments
- Attention to Detail – rigorous approach to coding, analysis, and documentation
- Leadership – mentoring peers and assisting in course development
- Time Management – effectively balancing multiple academic and research priorities
- Critical Thinking – evaluating data-driven insights and methodologies
- Creativity – innovative thinking in modeling and solution development

Work Experience

Teaching Assistant – Probability and Statistics

Indian Institute of Information Technology, Lucknow

Aug 2024 – Present

- Assisted in delivering undergraduate Probability course
- Evaluated assignments, quizzes, and examinations
- Supported enhancements to course material and structure

Projects

Detection of Distress Signals for Women’s Safety Using Image Processing

Designed a deep learning-based image classification system to detect emergency gestures and trigger alerts, achieving an accuracy of 91%. Paper under review.

Quantum Modeling of Financial Crashes

Developed models for early crash prediction using Quantum Boltzmann Machines, QGANs, and QNNs.

Stock Movement Prediction from Reddit Sentiment

Analyzed subreddit sentiment to forecast stock movements using NLP and deep learning, achieving an R^2 score of 0.95.

Fractal-Based Quantum-Enhanced Framework

Ongoing thesis on modeling complex systems using fractals and quantum computing tools.

Dissertation – Study on Poverty, Inequality, and Corruption

Explored interdependency using statistical models and socio-economic indicators.

Publications and Conferences

Presented: Statistical Analysis of War and Post-War Economics – 49th OMS Conference, 2022

Other Interests

Reading, Photography, Video Games, Personal Fitness